Finite covers by monic one-variable standard algebras

A shorter local proof and a direct construction for one equation

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The implication "finitely presented, flat and formally unramified implies formally étale" is checked in `sag/etale-direct`. This note studies the stronger conclusion: a finite principal cover by monic one-variable standard étale algebras over the original base. It removes the degree-lowering loop from the local proof, and gives a direct mathematical construction for one equation, including a bound on the number of charts. The shortened local argument is now checked in `sag/etale-monic-local`; the finite-cover construction remains mathematical. The final part gives the exact correspondence with the formalisation and distinguishes the remaining obligations.

Part I
Mathematics

1 The precise target

All rings are commutative and may be the zero ring. A monic standard étale algebra over R has the form

$$(R[T]/(P))[1/(G(T)P'(T))], \qquad P \text{ monic.}$$

The desired conclusion for an R-algebra E consists of finitely many $h_\alpha, c_\alpha \in E$, an identity

$$\sum_\alpha c_\alpha h_\alpha = 1,$$

and monic standard presentations of each $E[1/h_\alpha]$ over R. Both the identity and the actual algebra isomorphisms matter.

One cannot replace this by a single global generator. For example, $\mathbf{F}_2^3$ is étale over $\mathbf{F}_2$, but any element has at most two distinct coordinate values, and therefore cannot generate this algebra. Nor does covering the closed fibre over a local base suffice: for $R = k[t]_{(t)}$, the algebra $R \times R[1/t]$ has a whole component invisible in its closed fibre.

2 Surjective standard sources suffice

The recent direct lifting theorem removes an entire issue from the construction of the cover: once a standard source maps onto a proposed chart, its kernel can be removed afterwards.

Lemma 1 (Removing the kernel)**.** *Suppose that E is a finitely presented formally étale R-algebra, and there is a surjection*

$$u : S = (R[T]/(P))[1/(GP')] \longrightarrow E, \qquad P \text{ monic.}$$

Then for some $F \in R[T]$,

$$E \simeq (R[T]/(P))[1/(GFP')].$$

In particular the monic polynomial and its degree are unchanged.

Proof. Let $K = \ker u$. This ideal is finitely generated: choose preimages in S of the finitely many presentation generators of E; the defining relations of E and the equations recovering the generators of S give a finite list of kernel generators.

Formal smoothness of E lifts its identity to a section $E \to S/K^2$. The composite $S \to E \to S/K^2$ and the quotient map $S \to S/K^2$ agree modulo the square-zero ideal K/K^2. Formal unramifiedness of S makes these maps equal, whence $K = K^2$. The determinant trick for a finitely generated idempotent ideal gives $K = (e)$ with $e^2 = e$. Consequently

$$E = S/(e) \simeq S[1/(1-e)].$$

Represent $1 - e$ as $F(T)/(GP')^a$. Localising at it is the same as inverting $F(T)$ in S. □

This lemma has no residue-field or flatness hypothesis. In the original flat–unramified problem, its formal étaleness hypothesis is supplied by the checked direct theorem.

Proposition 2 (Reduction to square Jacobian algebras)**.** *For the original flat–unramified problem it suffices to construct monic standard covers of*

$$S = \big(R[X_1, \dots, X_r]/(F_1, \dots, F_r)\big)[1/\Delta], \qquad \Delta = \det(\partial F_i/\partial X_j).$$

Proof. The direct theorem constructs a surjection from such an S to E, and proves E formally étale. The proof of Lemma 1, through $K = (e)$, works for this S too. Thus $E \simeq S[1/(1-e)]$. Restrict a finite standard cover of S to this principal open. A further principal localisation of a standard algebra is standard: express its denominator as a polynomial in the standard generator divided by a power of the existing denominator. The unit-ideal identity maps from S to E. □

3 Removing the degree-lowering loop from the local proof

There is a substantial simplification even before attempting a direct construction of the whole cover. Once a localisation is generated by one integral element and a polynomial denominator, formal unramifiedness supplies a simple monic relation without any calculation in the special fibre.

Lemma 3 (Monicisation of a simple relation)**.** *Let E be a finitely presented formally unramified R-algebra. Suppose that $y \in E$ satisfies a monic polynomial $Q_0 \in R[T]$, that $G(y)$ is invertible, and that the map*

$$R[T, 1/G(T)] \longrightarrow E, \qquad T \longmapsto y,$$

is surjective. Then there is a monic polynomial Q with

$$Q(y) = 0, \qquad Q'(y) \in E^\times.$$

Consequently there is a surjection from a monic standard étale algebra onto E. If E is also formally smooth, it is itself monic standard étale over R.

Proof. The kernel of the displayed map is finitely generated. Clearing the denominators of finitely many generators gives polynomials $F_1, \dots, F_s \in R[T]$ such that

$$E = R[T, 1/G]/(F_1, \dots, F_s).$$

We first show that the elements $F_j'(y)$ generate the unit ideal in E. Put $C = E/(F_1'(y), \dots, F_s'(y))$. There are two R-algebra maps

$$E \rightrightarrows C[\varepsilon]/(\varepsilon^2), \qquad y \longmapsto \bar{y}, \quad y \longmapsto \bar{y} + \varepsilon.$$

The second is well defined because $F_j(\bar{y}+\varepsilon) = F_j(\bar{y}) + F_j'(\bar{y})\varepsilon = 0$, and $G(\bar{y}+\varepsilon)$ is invertible, being invertible modulo a square-zero ideal. The maps agree modulo ε. Formal unramifiedness makes them equal, so $\varepsilon = 0$, whence $1 = 0$ in C. Effectiveness of the ideal quotient gives

$$1 = \sum_j b_j F_j'(y), \qquad b_j \in E.$$

Express the finitely many b_j as polynomial fractions in y with common denominator $G(y)^N$. There are $A_j \in R[T]$ with

$$G(y)^N = \sum_j A_j(y) F_j'(y).$$

Set $H = \sum_j A_j F_j$. Then

$$H(y) = 0, \qquad H'(y) = G(y)^N.$$

Choose a coefficient list for H of length k, so that every recorded term of H has exponent less than k. Put

$$Q = H + T^k Q_0^2.$$

The two coefficient lists concatenate: the coefficients of H come first, followed by those of the monic polynomial Q_0^2. Thus Q is monic. Both Q_0^2 and its derivative vanish at y, so

$$Q(y) = 0, \qquad Q'(y) = H'(y) = G(y)^N,$$

and the latter is a unit. This construction also covers $Q_0 = 1$ and the zero algebra; no decision about a polynomial's actual degree is needed. The alternative $Q_0^m + H$, with m sufficiently large, works when Q_0 has positive degree. Thus

$$(R[T]/(Q))[1/(GQ')] \twoheadrightarrow E.$$

If E is formally smooth, apply Lemma 1. □

Proposition 4 (An interface before localisation)**.** *Let E be a finitely presented formally unramified R-algebra, $y \in E$ integral over R, and $G \in R[T]$. Suppose that for every $e \in E$ there merely exist $A \in R[T]$ and $n \geq 0$ such that*

$$A(y) = G(y)^n e.$$

If E is formally smooth, or if E is R-flat, then $E[1/G(y)]$ is merely a monic standard étale R-algebra.

Proof. Principal localisation preserves finite presentation, formal unramifiedness and formal smoothness. The displayed fraction generation and the integral equation for y give the hypotheses of Lemma 3 in the localisation. In the flat case the direct flat–unramified theorem first supplies formal smoothness of E. Every finite choice is made inside the truncation of the conclusion. In particular $G(y)$ need not be invertible in E. □

Application to the existing local proof. Immediately after its determinant argument, `zmt2-revised.tex` has

$$E_g = R[y][1/G(y)],$$

where y belongs to a finite R-subalgebra and is therefore integral. Lemma 3 applies at exactly this point. It replaces the construction of $\bar{H}$, the gcd/Nakayama degree-lowering loop, and the special-fibre comparison used to find a suitable monic relation. Its derivative is already invertible throughout E_g: no further shrinking by a derivative is needed on the target. For a flat unramified input, the checked direct theorem supplies formal smoothness, and kernel removal finishes.

This does not remove the earlier construction of the finite subalgebra, the residual monogenic factors, the residual idempotent, or the determinant that produces G. It also does not by itself turn the local charts into a finite cover over an arbitrary base. It does remove one of the recursive parts of the local candidate construction.

4 A cofactor lemma for recovering coordinates

The following calculation does not require a free basis for the algebra. The entries of v below need only be a finite list of elements.

Lemma 5 (Normalised eigenvector)**.** *Let $M \in M_n(R)$ and let $y, v_0, \dots, v_{n-1}$ belong to an R-algebra B, with $v_0 = 1$ and $Mv = yv$. Set*

$$P(T) = \det(TI - M), \qquad A(T) = \operatorname{adj}(TI - M), \qquad d_j(T) = A_{0j}(T).$$

Then

$$P(y) = 0, \tag{1}$$

$$A_{ij}(y) = v_i d_j(y), \tag{2}$$

$$P'(y) = \sum_{j=0}^{n-1} v_j d_j(y). \tag{3}$$

Consequently the opens at $P'(y)d_j(y)$ cover $B[1/P'(y)]$. On the jth such open,

$$v_i = A_{ij}(y)/d_j(y).$$

Proof. Write $L = yI - M$. The equation $Lv = 0$, with $v_0 = 1$, expresses column 0 of L as minus the sum of its other columns multiplied by the corresponding v_i. Multilinearity and alternation of the determinant give $\det L = 0$. Apply the same column substitution to the cofactor obtained by deleting row j and column i. Every term with a repeated column vanishes, and the remaining term, with its cofactor sign, is v_i times the cofactor obtained by deleting row j and column 0. This gives (2), without cancelling any determinant or assuming $P'(y)$ invertible.

Differentiating the determinant of $TI - M$ gives $P'(T) = \operatorname{tr} A(T)$, by multilinearity in the columns. Substitute (2) on the diagonal to obtain (3). After inverting $P'(y)$ this is a unit-ideal certificate for the $d_j(y)$. □

The needed matrix identities are therefore ordinary determinant multilinearity and alternation. There is no rank test, basis extraction, division by a nonunit, or appeal to prime ideals in this lemma.

5 A direct finite cover for one equation

Theorem 6. *Let*

$$f(X) = \sum_{r=0}^{n} a_r X^r \in R[X], \qquad E = (R[X]/(f))[1/f'(x)].$$

For $n \geq 2$, there is a finite cover of E by at most $2n^2$ monic standard étale algebras over R. Their monic polynomials have degree n. The integer n may be an upper bound for the degree: no nonvanishing assumption on a_n is required.

5.1 Horner tails and finite matrices

Put $D = f'(x)$, which is a unit in E, and define

$$t_i = a_i + a_{i+1}x + \cdots + a_n x^{n-i}, \qquad 1 \leq i \leq n.$$

Let V be the R-submodule spanned by $1, x, \ldots, x^{n-1}$. Both t_i and xt_i preserve V. More precisely, for the exponents needed here,

$$t_i x^j = \begin{cases} \displaystyle\sum_{r=i}^{n} a_r x^{r-i+j}, & j < i, \\ \displaystyle -\sum_{r=0}^{i-1} a_r x^{r+j-i}, & j \geq i. \end{cases} \tag{4}$$

The second identity follows by subtracting $x^{j-i}f(x) = 0$. For t_i use $0 \leq j \leq n-1$, and for xt_i use $1 \leq j \leq n$; every exponent on the right is at most $n-1$.

For each i and $c \in \{0, 1\}$ put

$$y_{ic} = (x + c)t_i.$$

Formula (4) gives a concrete matrix $M_{ic} \in M_n(R)$ such that

$$M_{ic}(1, x, \ldots, x^{n-1})^{\mathsf{t}} = y_{ic}(1, x, \ldots, x^{n-1})^{\mathsf{t}}.$$

Its row j is obtained by adding the coefficient row for $t_i x^{j+1}$ to c times the coefficient row for $t_i x^j$. Define

$$P_{ic}(T) = \det(TI - M_{ic}), \qquad \delta_{ic} = P'_{ic}(y_{ic}).$$

Each P_{ic} is monic of degree n and vanishes at y_{ic} by Lemma 5. The module V need not be free, nor need it equal E as an R-module.

5.2 The derivatives generate the unit ideal

Here is the finite algebraic replacement for selecting the degree of f over a residue field. Work modulo $(a_{i+1}, \ldots, a_n)$. The first i rows of M_{ic} have support in the first i columns. The remaining diagonal block is triangular with diagonal $-a_{i-1}$. The characteristic polynomial is consequently

$$P_{ic}(T) = (T + a_{i-1})^{n-i}\Phi_{ic}(T), \tag{5}$$

where the monic polynomial of the first block is

$$\Phi_{ic}(T) = (T - ca_i)^i + \sum_{r=0}^{i-1} a_r a_i^{i-1-r}(T - ca_i)^r.$$

This formula has only nonnegative exponents, including when $i = 1$. It is the usual companion-matrix determinant, with scaled variable. In this quotient $y_{ic} = a_i(x + c)$, and direct substitution gives

$$\Phi_{ic}(y_{ic}) = a_i^{i-1} f(x) = 0, \qquad a_i \Phi'_{ic}(y_{ic}) = a_i^{i-1} D.$$

Writing $w_c = a_i(x + c) + a_{i-1}$, differentiation of (5) therefore gives

$$a_i \delta_{ic} = a_i^{i-1} w_c^{n-i} D \pmod{(a_{i+1}, \ldots, a_n)E}. \tag{6}$$

In particular $w_1 - w_0 = a_i$.

Let J be the ideal generated by the $2n$ elements δ_{ic}, and use the constructive radical

$$\sqrt{J} = \{z : \text{ for some } N \geq 1,\ z^N \in J\}.$$

Its ideal laws follow from finite binomial expansions. We prove $a_i \in \sqrt{J}$ by descending induction. Once the higher coefficients belong to $\sqrt{J}$, equation (6) and invertibility of D give

$$a_i^{i-1} w_c^{n-i} \in \sqrt{J} \qquad (c = 0, 1).$$

For $i = n$, this gives $a_n^{n-1} \in \sqrt{J}$, hence $a_n \in \sqrt{J}$, because $n \geq 2$. For $1 < i < n$, it gives $a_i w_0, a_i w_1 \in \sqrt{J}$: raise each product to a sufficiently large power and use radicality. Subtracting gives $a_i^2 \in \sqrt{J}$, hence $a_i \in \sqrt{J}$. For $i = 1$, it gives $w_0, w_1 \in \sqrt{J}$, since $n - 1 \geq 1$; their difference is a_1.

Finally $D = \sum_{i=1}^{n} i a_i x^{i-1}$ belongs to $\sqrt{J}$. Since D is invertible, $1 \in \sqrt{J}$, and hence $1 \in J$. This argument uses only finite ideal and power witnesses. It contains no case distinction on whether a coefficient is zero or invertible.

5.3 Cofactor charts and standard sources

For a fixed pair (i, c) abbreviate $y = y_{ic}$, $P = P_{ic}$, $\delta = P'(y)$, and put

$$A(T) = \operatorname{adj}(TI - M_{ic}), \quad d_j(T) = A_{0j}(T), \quad p_j(T) = A_{1j}(T).$$

The row with index 1 exists because $n \geq 2$. Lemma 5 supplies the identities

$$d_j(y)x = p_j(y), \qquad \delta = \sum_{j=0}^{n-1} x^j d_j(y).$$

Set

$$h_{icj} = \delta_{ic} d_j(y_{ic}).$$

For each (i, c), δ_{ic}^2 belongs to the ideal generated by its h_{icj}. Since the δ_{ic} generate the unit ideal, so do all the h_{icj}. Explicitly, expand a sufficiently high power of a unit-ideal identity for the δ_{ic} so that every summand contains a square, and substitute the displayed cofactor identity. There are $2n^2$ indices (i, c, j); zero charts may be retained.

On $E[1/h_{icj}]$, both δ and $d_j(y)$ are invertible, and $x = p_j(y)/d_j(y)$. Define the following polynomial, with no fractions:

$$H_j(T) = \sum_{r=1}^{n} r a_r p_j(T)^{r-1} d_j(T)^{n-r}.$$

It satisfies

$$H_j(y) = d_j(y)^{n-1} D$$

on this chart, so it is invertible there. Thus evaluation at y gives a homomorphism from the monic standard algebra

$$S_{icj} = (R[T]/(P))[1/(P' d_j H_j)] \longrightarrow E[1/h_{icj}].$$

This homomorphism is surjective: it contains the generators

$$x = \frac{p_j(y)}{d_j(y)}, \qquad D^{-1} = \frac{d_j(y)^{n-1}}{H_j(y)}, \qquad h_{icj}^{-1} = \frac{1}{P'(y) d_j(y)}.$$

The target is finitely presented and formally étale. Indeed a simple root of f lifts uniquely across a square-zero ideal by the correction $a \mapsto a - f(a)/f'(a)$, and principal localisation preserves this property. Lemma 1 now turns the surjection into a standard presentation, multiplying its denominator by one further polynomial and keeping P unchanged. This proves Theorem 6.

5.4 Small bounds and a quadratic check

For $n = 0$, the inverted derivative is zero and E is the zero ring. For $n = 1$,

$$E \simeq R[1/a_1], \qquad x = -a_0/a_1,$$

which is standard with monic polynomial T and denominator a_1.

For $f = aX^2 + bX + c$, the choice $i = 2$, $c = 0$ in the construction gives

$$M = \begin{pmatrix} 0 & a \\ -c & -b \end{pmatrix}, \quad y = ax, \quad P(T) = T^2 + bT + ac, \quad \operatorname{adj}(TI - M) = \begin{pmatrix} T + b & a \\ -c & T \end{pmatrix}.$$

Here $P'(y) = 2ax + b = D$ is already a unit. The two cofactor denominators are $ax + b$ and a, and

$$D = (ax + b) + ax.$$

They give exactly the two opens in the direct quadratic example in `zmt2-revised.tex`. Thus the quadratic construction extends to arbitrary degree through finite matrices and a bounded induction.

6 What this does and does not settle

The construction proves the monic-cover theorem directly for a single equation with invertible derivative. It works over the original ring, including nonreduced rings and finite fields. It uses neither ZMT, residue-field factorisation, selected primes, nor a dynamical interpretation of the local proof.

For a general square Jacobian algebra the unresolved task is still to obtain such one-variable charts. Gaussian elimination on the Jacobian does not by itself eliminate polynomial variables. Similarly, replacing a system by a tower of one-variable extensions would still require an argument turning that tower into a finite monic standard cover. Monogenicity is not preserved by arbitrary compositions: over $\mathbf{F}_2$, the tower $\mathbf{F}_2 \to \mathbf{F}_2^2 \to \mathbf{F}_2^4$ already illustrates the problem.

Lemma 5 suggests a concrete algebraic interface for a possible extension. For each of finitely many candidates, provide:

1. a finite vector $v = (1, x_1, \ldots, x_r, \ldots)$ of elements of the target algebra;
2. a matrix M over R and an element y with $Mv = yv$;
3. a unit-ideal identity for the corresponding $\det(TI - M)'|_{T=y}$, over all the candidates.

The cofactor construction then expresses every chosen algebra generator as a rational function of y on finitely many opens. It provides surjective standard sources, and Lemma 1 finishes. Any localisation generators must also be included or have their inverses explicitly supplied, as D^{-1} was above.

This interface is sufficient, not an already proved existence theorem. In particular a finite spanning module with arbitrary multiplication matrices can give a characteristic polynomial with repeated roots. Formal unramifiedness of the target does not make the derivative of such an arbitrary characteristic polynomial invertible. The Horner construction above succeeds because the block calculation controls those extra factors and the coefficient induction covers their failures. An analogue of that calculation for several variables is the main research question.

Part II
Connection with the checked development

7 Already available

- `sag/etale-direct/FlatUnramifiedEtale.seg` proves the arbitrary-base lifting theorem from finite presentation, flatness and formal unramifiedness.
- `sag/etale-direct/UnramifiedEtaleSource.seg` constructs the surjective square Jacobian source. `JacobianLifting.seg` supplies the square-zero Newton correction.
- `sag/etale-local/EtaleCase.seg`, together with `IdempotentIdeal.seg`, proves the kernel-removal mechanism in Lemma 1. Its application needs finite kernel generators, fraction representatives and the explicit source presentation. These bridges are now constructed for the local shortcut in `sag/etale-monic-local`.
- `sag/etale-dynamical-global/StandardChartLocalisation.seg` proves closure of standard presentations under further principal localisation, with the algebra universal property.
- The determinant, adjugate, characteristic-polynomial, polynomial and localisation developments provide the underlying operations. In particular `Adjugate.seg` defines cofactors by column replacement and proves their linearity and adjugate identities.

These are reusable ingredients. They do not mean that the new finite cover or its unit-ideal certificate has already been checked.

8 The checked shortened local proof

The development `sag/etale-monic-local` proves Lemma 3 and Proposition 4. Its public endpoint is

> `ShortLocalEtale.Over[R,setR,cs].At[E,y,full,unique]`
> `theoremfpintegralsmooth`, or
> `flat-theoremfpintegralflat`.

Here G is the polynomial represented by `cs`; `full` is exactly the fraction-generation hypothesis of Proposition 4; and `unique` is formal unramifiedness. Both `fp` and `integral` are truncated: neither presentation generators nor the monic integral equation have to be chosen outside the conclusion.

The output merely consists of a localisation witness for $E[1/G(y)]$, including its algebra universal property, a coefficient list specifying a monic Q, a polynomial denominator D, and a standard presentation

$$E[1/G(y)] \simeq (R[T]/(Q))[1/(DQ')].$$

Standard presentations are expressed by the checked `StandardEtale.Points.Standard` universal property, not just by a surjective map or an equality of sets of points.

The construction has the following correspondence with the proof.

- `FiniteSurjectionKernel.seg` constructs finite kernel generators for a surjection between finitely presented algebras. It is used both for $R[T, 1/G] \to E$ and for the final standard source.
- `PolynomialLocalisation.seg` and `PolynomialFractions.seg` construct the source presentation, maps and fraction arithmetic.
- `MonicDualTest.seg` and `MonogenicDerivativeIdeal.seg` construct the dual-number map over the quotient by the derivatives and obtain $1 = \sum_j b_j F_j'(y)$. These witnesses are proved from unramifiedness, not added as hypotheses. This step uses the direct dual-number test rather than an application of `DiagonalJacobian.seg`.
- `SimplePolynomialRelation.seg` clears the finite family of fraction coefficients and proves $H(y) = 0$ and $H'(y) = G(y)^N$. `MonicRelation.seg` then constructs exactly $H + T^k Q_0^2$ by coefficient-list concatenation and proves its value and derivative.
- `MonogenicStandard.seg` builds the monic standard surjection, constructs its finite kernel and calls the unchanged `EtaleCase.seg` kernel-removal theorem.
- `PrincipalEtale.seg`, `LocalMonogenicStandard.seg` and `ShortLocalEtale.seg` provide localisation and the public interface above. The flat variant calls the unchanged `FlatUnramifiedEtale.seg` theorem.

No field, prime, ZMT, classical-logic or choice principle is a new hypothesis. The ambient univalence, replacement and surjection parameters are the same as in the reused development. The old local proof file is not rewritten: the new theorem supplies its replacement ending after the determinant and fraction-generation data have been constructed. The earlier construction of those data remains necessary.

A fresh run checks 6,795 declarations with zero failures in 2.78 seconds; 679 declarations in 12 proof modules are new. The opening audit covers all 657 named new declarations; each opens only its declared list, with at most 33 definitions opened and at most 0.01 seconds per new declaration in this run. The module check passes all 168 modules.

Regressions include repeated initial monic relations, redundant and empty presentations, arbitrary already-standard algebras, and localisations of $\mathbf{Z}/4\mathbf{Z}$ at 1, 2 and 0. A concrete two-denominator calculation checks the cleared exponent $1 + 2 = 3$. Mutations removing the square from Q_0^2, changing the tangent coefficient from 1 to 0, and shifting the cleared exponent are all rejected by the checker. These mutation checks use temporary copies; the development itself remains unchanged.

9 The separate finite-cover construction

The one-equation cover theorem has not been formalised in this work. It remains a separate, bounded target:

1. Prove the normalised-eigenvector cofactor identities and the trace formula for the derivative of the characteristic polynomial. Use the existing column conventions carefully: the mathematical matrices here are written with row index first, whereas Unit's determinant API stores columns first.

2. Construct the Horner multiplication matrices and their eigenvector equations. Prove the block determinant formula (5), including $i = 1$, zero leading coefficients and the empty lower block.

3. Convert the descending radical argument into finite power and ideal-combination certificates. This needs no prime-existence axiom and no choice of an actual degree for f.

4. Construct the $2n^2$ sources, prove surjectivity from the cofactor expressions, instantiate kernel removal, and export the finite cover with the original monic characteristic polynomials.

The new work is mainly the two determinant calculations and their polynomial bookkeeping. The previous lifting and idempotent-kernel proofs can be reused. A successful formalisation of this milestone would not yet complete the several-variable cover theorem.

As an informal arithmetic check, the matrix construction was tested on 2,136 characteristic-polynomial block identities and 806 instances with a root and an invertible derivative. The rings were $\mathbf{Z}/m\mathbf{Z}$ for $m \in \{2, 3, 4, 5, 6, 8\}$. The cofactor identities and the unit ideal generated by the proposed chart denominators passed. These finite tests support the calculations; they do not replace the proof above or a Unit check.

10 Relation to the local route

The existing local route remains an alternative for the general case: the field product theorem, local monogenic generation, finite coefficient-and-denominator descent, monic rescaling, and finite cover extraction. Lemma 3 now shortens the construction of the standard source; the direct lifting theorem supplies its formally smooth target. Neither automatically supplies the remaining global cover certificate.

For comparison, Stacks Proposition 10.144.4 gives the local standard-presentation theorem using a finite subalgebra and a primitive element in a residue field. The point of the present one-equation calculation is that this step can instead be performed through explicit matrices and ideal certificates in that case. Whether the same replacement can be made comparably short for a general square Jacobian system remains open here.